THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE UNIT TEST SET 2

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The area of a rectangle is 144 cm^2 and its length is 16 cm. Its breadth (in cm) is: 1

(a) 7 **(b)** 25 **(c)** 18 **(d)** 9

2. $1\frac{1}{2} + 2\frac{1}{3} =$ 1

(a) $3\frac{5}{6}$ **(b)** $3\frac{1}{2}$ **(c)** $3\frac{2}{5}$ **(d)** $3\frac{1}{6}$

3. In rectangle ABCD, diagonals meet at O and $\angle AOB = 60^\circ$. Then $\angle OAB$ is (degrees): 1

(a) 120 **(b)** 70 **(c)** 60 **(d)** 30

4. A figure coincides with itself after turns of 90, 180, 270, ... degrees, up to a full turn of 360°. Its order of rotational symmetry is: 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 6

5. The number of integers between -5 and 4 is: 1

(a) 10 **(b)** 9 **(c)** 8 **(d)** 7

6. Fencing a rectangular plot of perimeter 80 m costs ₹15 per metre. The total cost is (₹): 1

(a) 1215 **(b)** 1300 **(c)** 1185 **(d)** 1200

7. Arrange in ascending order: $\frac{3}{4}, \frac{2}{3}, \frac{5}{6}$. 1

(a) $\frac{3}{4}, \frac{2}{3}, \frac{5}{6}$ **(b)** $\frac{3}{4}, \frac{5}{6}, \frac{2}{3}$
(c) $\frac{2}{3}, \frac{3}{4}, \frac{5}{6}$ **(d)** $\frac{2}{3}, \frac{5}{6}, \frac{3}{4}$

8. A circle of radius 3 cm and a concentric circle of radius 7 cm are drawn. The width of the ring between them (cm): 1

(a) 6 **(b)** 3 **(c)** 4 **(d)** 5

9. A regular polygon has 7 lines of symmetry. The number of its sides is: 1

(a) 6 **(b)** 8 **(c)** 7 **(d)** 9

10. A diver at -15 m (below sea level) swims up 6 m. His new position (m) is: 1

(a) -11 **(b)** -7 **(c)** -9 **(d)** 3

11. How many square tiles of side 20 cm are needed to cover a floor of 400 cm by 300 cm? 1

(a) 6000 **(b)** 300 **(c)** 70 **(d)** 35

12. A pizza is cut into 12 equal slices. Riya eats 4 slices. The fraction she eats in simplest form has denominator: 1

(a) 5 **(b)** 3 **(c)** 4 **(d)** 2

13. The diagonals of a square are: 1

(a) equal but not perpendicular
(b) parallel
(c) unequal
(d) equal and perpendicular bisectors of each other

14. Which of these letters has rotational symmetry of order 2 but NO line of symmetry? 1

(a) H **(b)** A **(c)** Z **(d)** O

15. The sum of an integer and its opposite is always: 1

(a) 1 **(b)** twice the integer
(c) the integer itself **(d)** 0

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A rectangle of length 12 cm and breadth 5 cm is cut along a diagonal. Find the area of each triangle. 2

17. Subtract $\frac{1}{3}$ from $\frac{5}{6}$. 2

18. In rectangle ABCD the diagonals meet at O. If AC = 12 cm, find OA, OB, BD. 2

19. Find the angle of rotational symmetry and order for a regular polygon of 5 sides. 2

20. Compare -4 and -7 using <, > or =. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In a square with diagonal intersection O and side 4 cm, state the angle between the diagonals and the angle $\angle OAB$. 3

22. For the capital letter A, state whether it has 3
(i) a vertical line of symmetry,
(ii) a horizontal line of symmetry,
(iii) rotational symmetry of order 2.

23. The temperature at 6 am is -4°C. It rises by 9°C by noon and then falls by 8°C by evening. Find the evening temperature. Is it below 0°C? 3

24. A square tile of side 20 cm is used to cover a floor of 4 m by 3 m. Find the number of tiles. (1 m = 100 cm) 3

OR

Find the missing number: $\frac{3}{5} = \frac{?}{20} = \frac{9}{?}$ 3

25. A circle has radius 4 cm. Find the lengths of its diameter and of its longest chord. Can there be a chord 9 cm long? 3

OR

A point P is 3 units above a horizontal mirror line. Where is its image and how far apart are P and its image? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A submarine is at -80 m. It rises 30 m, then dives 25 m. Find its final depth and the total vertical distance travelled. Is it below sea level? 5

OR

A rectangular garden is 20 m by 15 m. A path 2 m wide runs around it on the outside. Find the area of the path. 5

27. A tank holds $2\frac{1}{2}$ litres of water. $1\frac{1}{3}$ litres more are poured in and $\frac{1}{4}$ litre is used. Find the water left. 5

OR

A circle has centre O and radius 6 cm. Points P, Q, R are at distances 4 cm, 6 cm and 9 cm from O. State whether each lies inside, on or outside the circle, and how many lie inside or on the circle. 5

END OF QUESTION PAPER — Total: 50 Marks

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